

ROS 2 Quick Command Guide

Kiwi Drive Robot - Raspberry Pi / ESP32 Integration

Important: Run these commands in every new terminal before using the workspace:

Source ROS 2 + Workspace

```
source /opt/ros/lyrical/setup.bash
cd ~/robot_ws
source install/setup.bash
```

1. Build the ROS 2 Workspace

COMMAND	Build the complete ROS 2 workspace.
PURPOSE	Use this after changing ROS 2 packages, Python nodes, launch files, URDF files, configuration files, or package dependencies.
TERMINALS NEEDED	1 Terminal

Terminal 1

```
cd ~/robot_ws
colcon build --symlink-install
source install/setup.bash

# Build only one package:
colcon build --symlink-install --packages-select PACKAGE_NAME

# Example:
colcon build --symlink-install --packages-select kiwi_drive_description
```

2. View the Robot Model in RViz

COMMAND	Launch the Kiwi Drive robot model in RViz.
PURPOSE	Displays the robot URDF and TF frames so the robot model, joints, links, and coordinate frames can be inspected.
TERMINALS NEEDED	1 Terminal

Terminal 1

```
source /opt/ros/lyrical/setup.bash
cd ~/robot_ws
source install/setup.bash

ros2 launch kiwi_drive_description display.launch.py
```

3. Start the LD19 LiDAR

COMMAND	Start the LD19 LiDAR ROS 2 driver.
PURPOSE	Starts the LiDAR and begins publishing laser scan data to the /scan topic.
TERMINALS NEEDED	1 Terminal

Terminal 1

```
source /opt/ros/lyrical/setup.bash
cd ~/robot_ws
source install/setup.bash

ros2 launch ldlidar_ros2 ld19.launch.py
```

4. Start the LiDAR and View Live Scan Data

COMMAND Run the LD19 LiDAR and view the raw /scan messages.

PURPOSE Confirms that the LiDAR driver is running and that actual laser-distance measurements are being published.

**TERMINALS
NEEDED** 2 Terminals

Terminal 1 - Start LiDAR

```
source /opt/ros/lyrical/setup.bash
cd ~/robot_ws
source install/setup.bash

ros2 launch ldlidar_ros2 ld19.launch.py
```

Terminal 2 - View Scan Data

```
source /opt/ros/lyrical/setup.bash
cd ~/robot_ws
source install/setup.bash

ros2 topic echo /scan
```

5. Check the LiDAR Publishing Rate

COMMAND Measure how frequently the LiDAR publishes /scan messages.

PURPOSE Useful for checking whether the LiDAR is publishing at the expected rate and for diagnosing performance problems.

**TERMINALS
NEEDED** 2 Terminals

Terminal 1 - Start LiDAR

```
source /opt/ros/lyrical/setup.bash
cd ~/robot_ws
source install/setup.bash

ros2 launch ldlidar_ros2 ld19.launch.py
```

Terminal 2 - Check /scan Rate

```
source /opt/ros/lyrical/setup.bash
cd ~/robot_ws
source install/setup.bash

ros2 topic hz /scan
```

6. List Available ROS 2 Topics

COMMAND	List all currently available ROS 2 topics.
PURPOSE	Use this to see what information is currently being published or subscribed to in the ROS 2 system.
TERMINALS NEEDED	1 Terminal

Terminal 1

```
ros2 topic list

Common topics:
/cmd_vel
/scan
/odom
/tf
/tf_static
```

7. List Running ROS 2 Nodes

COMMAND	List all currently running ROS 2 nodes.
PURPOSE	Useful for checking whether expected nodes such as the serial bridge, LiDAR driver, robot_state_publisher, or odometry nodes are actually running.
TERMINALS NEEDED	1 Terminal

Terminal 1

```
ros2 node list
```

8. Check Information About a ROS 2 Topic

COMMAND	Display information about a ROS 2 topic.
PURPOSE	Shows the topic message type and the number of publishers and subscribers.
TERMINALS NEEDED	1 Terminal

Terminal 1

```
ros2 topic info /cmd_vel
ros2 topic info /scan
ros2 topic info /odom
```

9. Move the Robot Forward

COMMAND	Continuously publish a forward velocity command to /cmd_vel.
PURPOSE	Moves the robot forward using ROS 2 autonomous control. The ESP32 must be in AUTO mode and the serial bridge must already be running.
TERMINALS NEEDED	2 Terminals

Terminal 1 - Serial Bridge

```
source /opt/ros/lyrical/setup.bash
cd ~/robot_ws
source install/setup.bash

# Start the Kiwi serial bridge here.
```

Terminal 2 - Forward Command

```
source /opt/ros/lyrical/setup.bash
cd ~/robot_ws
source install/setup.bash

ros2 topic pub -r 10 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.3, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"

# Stop publishing with Ctrl+C
```

10. Move the Robot Backward

COMMAND Continuously publish a backward velocity command to /cmd_vel.

PURPOSE Moves the robot backward through the ROS 2 autonomous control path.

TERMINALS NEEDED 2 Terminals

Terminal 1 - Serial Bridge

```
# Run the Kiwi serial bridge.
```

Terminal 2 - Backward Command

```
ros2 topic pub -r 10 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: -0.3, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

11. Move the Robot Left

COMMAND Continuously publish a leftward holonomic velocity command.

PURPOSE Moves the robot sideways to the left without rotating first.

TERMINALS NEEDED 2 Terminals

Terminal 1 - Serial Bridge

```
# Run the Kiwi serial bridge.
```

Terminal 2 - Left Command

```
ros2 topic pub -r 10 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.3, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

12. Move the Robot Right

COMMAND Continuously publish a rightward holonomic velocity command.

PURPOSE Moves the robot sideways to the right without changing its orientation.

TERMINALS NEEDED 2 Terminals

Terminal 1 - Serial Bridge

```
# Run the Kiwi serial bridge.
```

Terminal 2 - Right Command

```
ros2 topic pub -r 10 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: -0.3, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

13. Rotate Counterclockwise

COMMAND Continuously publish a positive angular velocity command.

PURPOSE Rotates the robot counterclockwise in place.

TERMINALS NEEDED 2 Terminals

Terminal 1 - Serial Bridge

```
# Run the Kiwi serial bridge.
```

Terminal 2 - Rotation Command

```
ros2 topic pub -r 10 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.5}}"
```

14. Rotate Clockwise

COMMAND Continuously publish a negative angular velocity command.

PURPOSE Rotates the robot clockwise in place.

TERMINALS NEEDED 2 Terminals

Terminal 1 - Serial Bridge

```
# Run the Kiwi serial bridge.
```

Terminal 2 - Rotation Command

```
ros2 topic pub -r 10 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: -0.5}}"
```

15. Move the Robot Diagonally

COMMAND Publish X and Y translation commands at the same time.

PURPOSE Demonstrates holonomic movement by allowing the robot to move diagonally without first rotating.

TERMINALS NEEDED 2 Terminals

Terminal 1 - Serial Bridge

```
# Run the Kiwi serial bridge.
```

Terminal 2 - Diagonal Command

```
ros2 topic pub -r 10 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.3, y: 0.3, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

16. Move and Rotate at the Same Time

COMMAND	Publish translation and rotation simultaneously.
PURPOSE	Demonstrates full holonomic motion by moving the robot while also rotating it.
TERMINALS NEEDED	2 Terminals

Terminal 1 - Serial Bridge

```
# Run the Kiwi serial bridge.
```

Terminal 2 - Translate + Rotate

```
ros2 topic pub -r 10 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.3, y: 0.2, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.3}}"
```

17. Stop the Robot

COMMAND	Publish a zero-velocity command.
PURPOSE	Commands all translational and rotational velocities to zero.
TERMINALS NEEDED	1 Terminal if the serial bridge is already running

Terminal 1

First stop any repeating `ros2 topic pub` command with `Ctrl+C`.

Then run:

```
ros2 topic pub --once /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

18. View /cmd_vel Commands

COMMAND	Echo the <code>/cmd_vel</code> topic while another node or terminal publishes movement commands.
PURPOSE	Lets you verify exactly what X, Y, and rotational velocity commands are being sent through ROS 2.
TERMINALS NEEDED	2 Terminals

Terminal 1 - Publish Movement

```
ros2 topic pub -r 10 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.3, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

Terminal 2 - View /cmd_vel

```
ros2 topic echo /cmd_vel
```

19. View Robot Odometry

COMMAND	Echo the /odom topic.
PURPOSE	Displays the robot's estimated position, orientation, linear velocity, and angular velocity.
TERMINALS NEEDED	2 Terminals

Terminal 1 - Serial Bridge / Odometry Node

```
# Run the node that receives ESP32 odometry telemetry and publishes /odom.
```

Terminal 2 - View Odometry

```
source /opt/ros/lyrical/setup.bash
cd ~/robot_ws
source install/setup.bash

ros2 topic echo /odom
```

20. Check Odometry Publishing Rate

COMMAND	Measure how frequently /odom messages are being published.
PURPOSE	Useful for confirming that odometry feedback is updating at the expected rate.
TERMINALS NEEDED	2 Terminals

Terminal 1 - Serial Bridge / Odometry Node

```
# Run the Kiwi serial bridge / odometry node.
```

Terminal 2 - Check /odom Rate

```
ros2 topic hz /odom
```

21. View TF Data

COMMAND	View ROS 2 transforms.
PURPOSE	Used to inspect coordinate-frame relationships such as base_link, odom, LiDAR frames, and other robot links.
TERMINALS NEEDED	1 Terminal

Terminal 1

```
Dynamic transforms:
ros2 topic echo /tf

Static transforms:
ros2 topic echo /tf_static
```

22. View the Robot and LiDAR Together in RViz

COMMAND	Run the LiDAR driver and robot visualization simultaneously.
PURPOSE	Allows the robot model and LiDAR scan data to be viewed together in RViz.

TERMINALS NEEDED	2 Terminals minimum
Terminal 1 - LiDAR	
<pre>source /opt/ros/lyrical/setup.bash cd ~/robot_ws source install/setup.bash ros2 launch ldliidar_ros2 ld19.launch.py</pre>	
Terminal 2 - RViz / Robot Model	
<pre>source /opt/ros/lyrical/setup.bash cd ~/robot_ws source install/setup.bash ros2 launch kiwi_drive_description display.launch.py # In RViz, add a LaserScan display and set the topic to /scan.</pre>	

23. Full Robot Control Test

COMMAND	Run the serial bridge, LiDAR, RViz, and a movement command at the same time.
PURPOSE	Provides a complete system-level test of Raspberry Pi communication, ROS 2 control, LiDAR, visualization, and robot movement.
TERMINALS NEEDED	4 Terminals
Terminal 1 - Serial Bridge	
<pre>source /opt/ros/lyrical/setup.bash cd ~/robot_ws source install/setup.bash # Run the Kiwi serial bridge.</pre>	
Terminal 2 - LiDAR	
<pre>source /opt/ros/lyrical/setup.bash cd ~/robot_ws source install/setup.bash ros2 launch ldliidar_ros2 ld19.launch.py</pre>	
Terminal 3 - RViz	
<pre>source /opt/ros/lyrical/setup.bash cd ~/robot_ws source install/setup.bash ros2 launch kiwi_drive_description display.launch.py</pre>	
Terminal 4 - Movement	
<pre>source /opt/ros/lyrical/setup.bash cd ~/robot_ws source install/setup.bash ros2 topic pub -r 10 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.3, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"</pre>	

24. ESP32 Direct Serial Commands

COMMAND	Send commands directly to the ESP32 serial protocol.
PURPOSE	Useful for testing the ESP32 control layer without going through the full ROS 2 command path.
TERMINALS NEEDED	1 Serial terminal

Commands

```
MODE,AUTO
# Switch ESP32 to autonomous mode.

MODE,MANUAL
# Return control to the Bluetooth controller.

V,x,y,rotation
# Send a normalized movement command.
# Example: V,0.0,1.0,0.0

STOP
# Stop all motors.

PING
# Test serial communication.
# Expected response: ACK,PONG
```

25. Check Connected USB Serial Devices

COMMAND	List USB serial devices.
PURPOSE	Useful for identifying which /dev/ttyUSB port belongs to the ESP32 and which belongs to the LiDAR.
TERMINALS NEEDED	1 Terminal

Terminal 1

```
ls /dev/ttyUSB*

Example:
/dev/ttyUSB0
/dev/ttyUSB1
```

26. Quick Troubleshooting Commands

COMMAND	Use common ROS 2 diagnostic commands.
PURPOSE	These commands help identify missing packages, missing nodes/topics, LiDAR problems, odometry issues, and USB device problems.
TERMINALS NEEDED	1 Terminal for each check

Common Checks

```
# Rebuild and re-source workspace
cd ~/robot_ws
colcon build --symlink-install
source install/setup.bash

# Topics
ros2 topic list

# Nodes
ros2 node list

# LiDAR data
ros2 topic echo /scan

# LiDAR rate
ros2 topic hz /scan

# Velocity commands
ros2 topic echo /cmd_vel

# Odometry
ros2 topic echo /odom

# Odometry rate
ros2 topic hz /odom

# USB serial devices
ls /dev/ttyUSB*
```

Important Notes

1. Source ROS 2 in Every New Terminal

Commands

```
source /opt/ros/lyrical/setup.bash
cd ~/robot_ws
source install/setup.bash
```

2. ROS Movement Requires AUTO Mode

The ESP32 must be in AUTO mode before autonomous /cmd_vel commands are accepted.

3. Continuous Movement Commands

Commands using ros2 topic pub -r 10 continue publishing until Ctrl+C is pressed.

4. Autonomous Command Watchdog

The ESP32 expects repeated autonomous velocity commands. If the Raspberry Pi or ROS 2 stops sending commands for too long, the ESP32 automatically stops all motors.

5. Raspberry Pi / ROS 2 Responsibilities

LiDAR, SLAM, navigation, /cmd_vel generation, odometry publishing, and high-level autonomous control.

6. ESP32 Responsibilities

Xbox controller, manual/autonomous mode selection, Kiwi-drive inverse kinematics, encoder interrupts, wheel RPM measurement, velocity PID, motor PWM/direction, emergency stop, autonomous watchdog, and odometry telemetry.

7. Normal Command Flow

Command Flow

```
ROS 2 /cmd_vel
|
v
Serial Bridge
|
v
ESP32
|
v
driveKiwi()
|
v
Target Wheel RPM
|
v
Velocity PID
|
v
Motor PWM
```

8. Feedback Flow

Feedback Flow

```
Motor Encoders
|
v
ESP32
|
v
Measured Wheel RPM + Encoder Counts
|
v
ODOM Serial Telemetry
|
v
Raspberry Pi
|
v
ROS 2 /odom
```